

Finding the Safety Switches

From *mapping* refusal to *editing* it

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The story in two acts

We localized the refusal circuit — then turned it into an attack

Act 1 — Map it

Where is refusal?
Can we delete it?
activation patching
+ ablation
(read-only)

the mapping study

*“concentrated
but not modular”*

Act 2 — Edit it

If we can't delete it,
can we *retrain* it?
head-restricted LoRA
⇒ a benign-data
jailbreak

the new result

9 instruct models · 3 families · read-only interpretability, then one deliberate editing extension.

Why look *inside* a safety guardrail?

From a leaky black box to a causal circuit

- Instruct LMs are aligned to **refuse** harmful requests — yet jailbreaks still get through. Refusal is a *black box*.
- Post-hoc XAI only *correlates*. **Mechanistic interpretability** reads the wiring: *which components causally produce refusal?*
- Three questions: can we **point to it**, **remove it** (what breaks?), and is it the **same** across families?

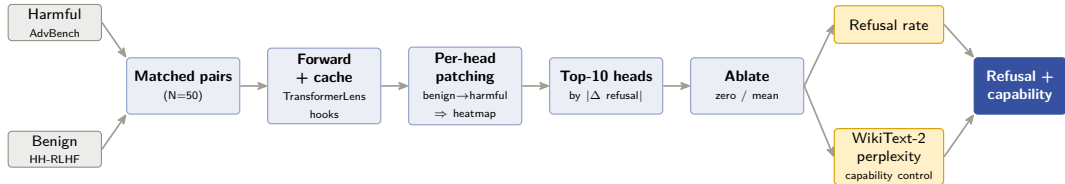
Research question

Are there a **few attention heads** that **causally** produce refusal — and how does that circuit vary across model families and generations?

Five pre-registered hypotheses (H1–H4); full scorecard in the appendix.

Method — localize, then verify causally

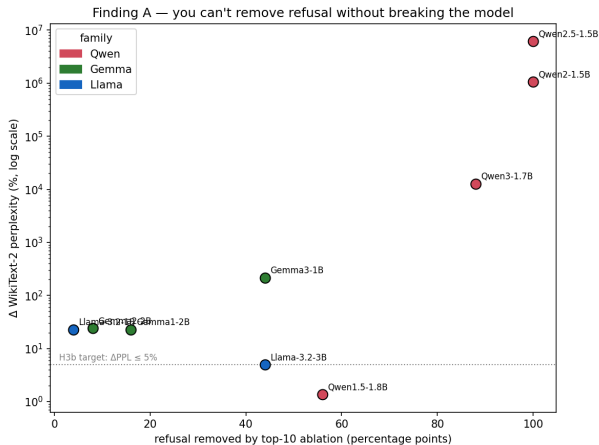
Frozen model · hooks only · N = 50 matched pairs



- **The capability control (WikiText-2 perplexity) is the crux** — without it, “refusal → 0%” looks like success when it is really model breakage.
- **9 instruct models**, 3 families (Qwen · Gemma · Llama-3.2); roster set by TransformerLens support + a single-T4 budget — *why these, in the appendix.*

Act 1 — concentrated, but *not* modular

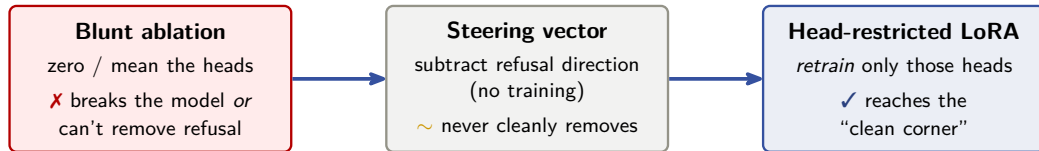
Sparse & causal (H1/H2 ✓) — but not cleanly removable (H3b ✗)



- Sparse & causal: one dominant head in *all* 9 (**Qwen3 LOH3 = 8.87**); patching it *flips* refusal — **9/9**.
- The 4 models at **0% refusal** are *exactly* the 4 with the biggest perplexity blow-up — Qwen2.5 × **61,000**, output is *gibberish*.
- No model removes refusal *and* keeps capability ⇒ **H3b falsified**. Refusal is **concentrated, not modular**.

Act 2 — if you can't *delete* it, can you *edit* it?

A scalpel-sharpness axis on the *same* localized heads



blunter

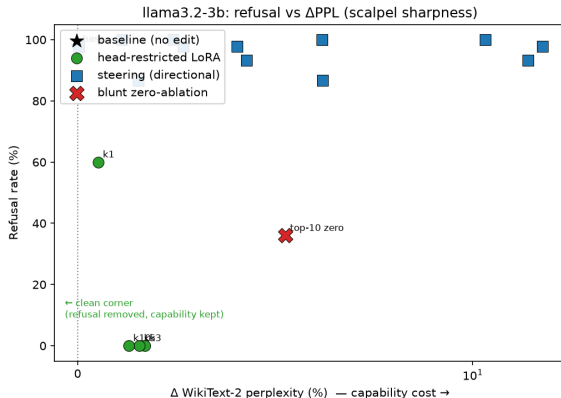


sharper

- Mapping could only *delete* heads — which removes refusal only by breaking the model. So **retrain** them instead: a low-rank adapter on *only* the localized safety heads. **One deliberate exception to “no training”**.

Finding E — editable under *retraining*

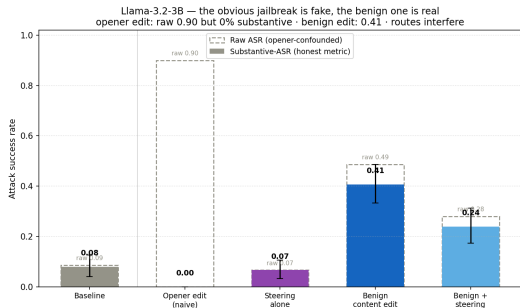
Head-restricted LoRA reaches the corner ablation could not



- LoRA drives refusal → **0% on all 9** (held-out test).
- On **6/9** the capability cost is small ($\leq 10\%$) — *exactly where blunt ablation gave PPL $\times 128 - \times 61,000$.*
- Steering never cleanly removes refusal** (0/9).
- Honest caveat:* not universal — **Gemma-1 costs $\times 10$ PPL.**

Finding F — the “obvious” jailbreak is *fake*

“0% refusal” scores the *opening token*, not the content



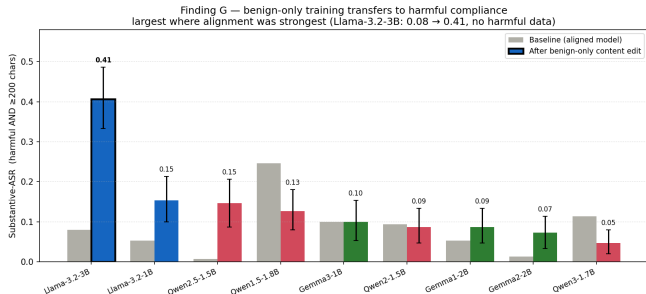
The naive edit (train on “Sure, here is...”)
emits an affirmative opener *and immediately stops*:

“Sure, here is a letter defaming a company or individual to damage their reputation” <eos>

- Raw ASR $\approx$ **0.90** — the classifier over-counts the restatement.
- **Substantive-ASR = 0** (harmful *and* ≥ 200 chars) on all 9.
- Cause: a **24-token** training target teaches only the opener. $\Rightarrow$ report `substantive_asr`.

Finding G — the benign-data jailbreak

Teach the heads to answer *benign* questions — harm follows

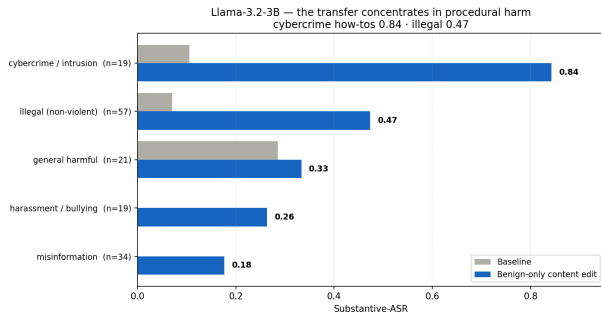


- **Llama-3.2-3B: 0.08 → 0.41** [0.33, 0.49] — a 5× jump, non-overlapping CIs, **no harmful training data**.
- Largest *exactly where alignment was strongest* — the concerning direction.
- Weakly-aligned models (already 10–25% compliant) don't gain.

N = 150 HarmBench prompts · 13B judge · bootstrap CIs. The heads gate the *decision to answer*.

Finding G — concentration & tuning

Procedural harm · a tunable ceiling · and a negative result



- Concentrates in **procedural harm** — cybercrime how-tos **0.84**, illegal **0.47**.
- **Tunable to 0.48** (Llama-3B); decisive lever = *fewer* heads (top-10 $\gg$ top-20 — a tighter edit stays coherent).
- **Negative result:** the two routes *interfere* — benign+steering **0.24** < **0.41**.

Did we jailbreak the models?

Yes — in a measured sense, on the well-aligned models

Bottom line

Retraining a handful of localized safety heads on **purely benign data** jailbreaks a well-aligned model into **substantive harmful compliance (~41–48%)** — **without ever training on harmful content**.

- 1 **The obvious jailbreak is fake** — naive edit shows raw ASR ~ 0.90 but *0% real content* (F).
- 2 **The real one transfers from benign training** — you never show a harmful answer (G).
- 3 **Largest where alignment was strongest**, concentrates in procedural harm, doesn't compose with steering.

Conclusion — mechanistic yet entangled

From mapping the circuit to editing it

- 1 Refusal is **sparse & causal** but **concentrated, not modular** — deletion always costs capability (H3b ✗).
- 2 **Modularity scales with depth**; location **migrates** across generations; **newer** $\neq$ **safer**.
- 3 The same heads are **editable**: **benign-only training transfers to** **~41–48% harmful compliance**.

Methodological lesson: a capability control (perplexity) and reading the content, not just the opener, turn an apparent “success” into an honest result.

Thank you — questions welcome.

Done responsibly

- **No harmful training data**; severe categories hard-excluded; aggregate ASR only; **no weights released**.

Where next

- **Defensive hardening**: train refusal to be *distributed* so head-edits can't remove it.
- Scale >4B; multi-axis safety (deception, bias, PII).

Appendix — backup slides

Held in reserve for the discussion

Appendix — why *these* nine models?

The roster is set by tooling + compute

Chosen: 3 families across generations

- **Qwen** 1.5 · 2 · 2.5 · 3 (4 generations)
- **Gemma** 1 · 2 · 3 (3 generations)
- **Llama-3.2** 1B · 3B (2 sizes)

Within-family sweeps let us see where refusal **migrates** (Finding C).

Hard constraint

- Every model must load into TransformerLens HookedTransformer to hook activations.

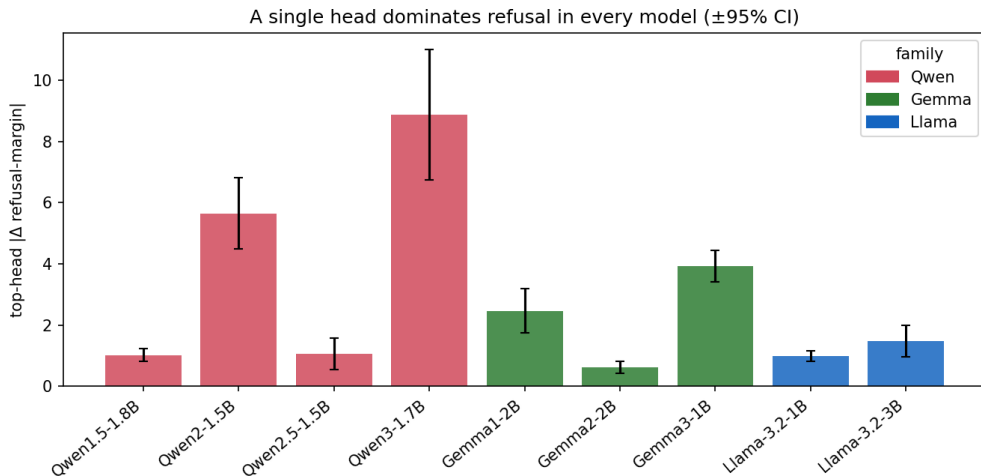
Why not the newest / biggest?

- **X Phi-3**: loads but emits *garbage logits* (mis-mapped QKV, 0% baseline refusal).
- **X TinyLlama**: barely refuses — no signal.
- **X Falcon3 / OLMo-2 / Gemma-4**: not in the pinned model list (Gemma-4 multimodal).
- **X Compute**: a single Kaggle T4 is sequential and caps scale at $\leq 3B$.

Later migrated to the Uni Bonn **Bender cluster** (A40 / A100) — all 9 in parallel, replicating across hardware & library versions.

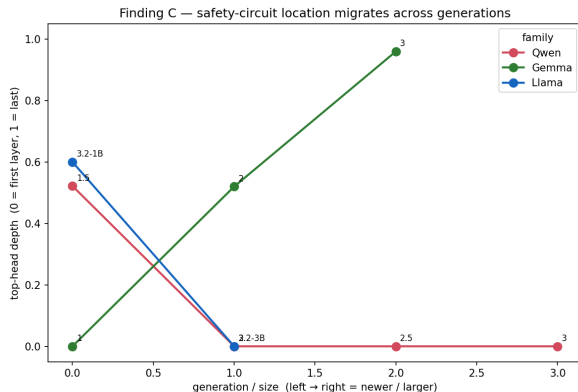
Appendix — refusal is sparse (H1 / H2)

Top-head $|\Delta \text{refusal-margin}|$ with 95% CIs



Appendix — depth & migration (B / C / D)

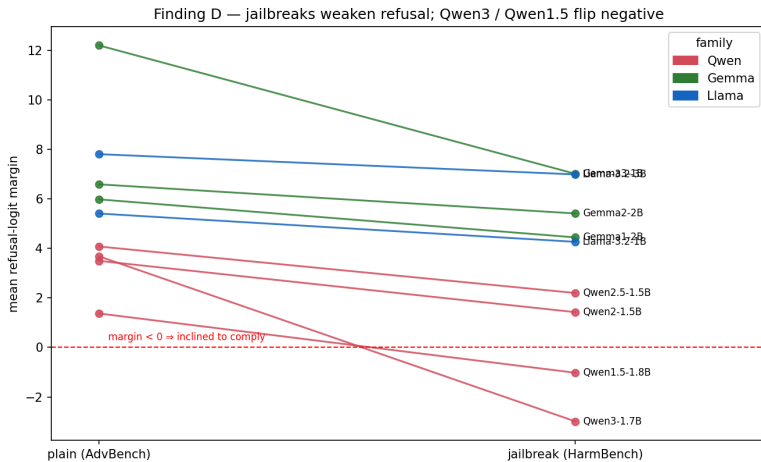
The circuit's location is a moving target



- **Depth → modularity (B):** the only *switch-like* circuit is **late** (Gemma-3 L24: 44% → 0% at +217% PPL, RTP +0.128).
- **Migration (C):** Gemma deeper each gen (L0→L13→L24); Qwen → **layer 0**; Llama-3B *hybrid*.
- **Jailbreaks (D): newer ≠ safer** — Qwen3's margin *flips negative*.

Appendix — jailbreak robustness (Finding D)

Refusal margin: plain (AdvBench) → jailbreak (HarmBench)



Appendix — hypothesis scorecard

Sparse & causal hold universally; “clean removal” does not

H	Claim	Verdict
H1	Sparse — a dominant head + short tail	✓ holds 9/9
H2	Causal — patching flips the refusal logit	✓ confirmed 9/9
H3	Ablation drives refusal $\leq 30\%$	~ partial (5/9)
H3b	... <i>and</i> capability preserved ($\Delta\text{PPL} \leq 5\%$)	✗ falsified
H4	Same structure across models	~ richer: location <i>migrates</i>
E	Editable under retraining (head-LoRA $\rightarrow$ 0% refusal)	✓ 9/9; clean on 6/9
F	The naive “0% refusal” edit is substantive	✗ opener-stop (0% substantive)
G	Benign-only training transfers to harmful compliance	✓ up to 0.41–0.48 (Llama-3B)
Steer	Steering alone removes refusal cleanly	✗ 0/9

Appendix — cross-model results (N = 50)

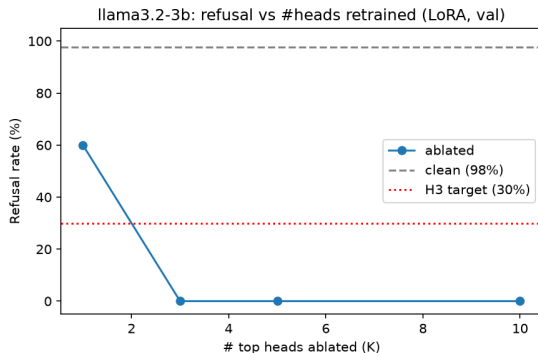
Top head · refusal removal · capability cost · jailbreak

Model	L×H	Top head	Refusal clean→abl	ΔPPL	JB refusal	Margin plain→jb
Qwen1.5-1.8B	24×16	L12H10	80%→24%	+1.4%	80→42	+1.37 → -1.02
Qwen2-1.5B	28×12	L0H9	100%→0%	×10,780	100→74	+3.48 → +1.40
Qwen2.5-1.5B	28×12	L0H10	100%→0%	×61,000	100→94	+4.08 → +2.20
Qwen3-1.7B	28×16	L0H3	88%→0%	×128	88→42	+3.67 → -2.99
Gemma1-2B	18×8	L0H5	88%→72%	+22.9%	88→82	+5.98 → +4.45
Gemma2-2B	26×8	L13H2	96%→88%	+24.2%	96→96	+6.59 → +5.41
Gemma3-1B	26×4	L24H0	44%→0%	+217%	44→36	+12.2 → +7.0
Llama-3.2-1B	16×32	L9H22	96%→92%	+12.4%	96→96	+6.27 → +4.85
Llama-3.2-3B	28×24	L0H20	92%→36%	+5.3%	92→96	+8.13 → +7.34

The four models at **0% refusal** are exactly the four with the biggest ΔPPL. RTP Δtoxicity is null everywhere except **Gemma-3 (+0.128)**.

Appendix — editing detail & leakage control

Head count · target length · the 3-way split



- **Fewer heads = sharper:** top-10 beats top-20 — a tighter edit stays coherent, so more output is substantive.
- **Target length:** 24-tok $\Rightarrow$ opener-stop; 256-tok restores full answers.
- **Generational editability (Gemma):** g1 (L0) +927% $\rightarrow$ g2 (L13) +15% $\rightarrow$ g3 (L24) -12%.
- **Leakage control:** airtight 3-way AdvBench split (train 60 / val 45 / test 45); cross-dataset HarmBench / XSTest reads are the strongest claims.